

Mahak Singh

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Summary

Computer Science undergraduate with hands-on experience in Java, Spring Boot, and REST API development through academic and internship projects, and exposure to MySQL. Comfortable working with Data Structures and Algorithms. Looking to start my career as a Software Development Engineer with a focus on backend development.

Education

B.Tech in Computer Science
ABES Engineering College

Oct 2023 – May 2027
CGPA: 7.7/10

Technical Skills

- **Programming Languages:** Java (Core Java, OOP)
- **Backend:** Spring Boot, REST APIs
- **Database:** SQL (MySQL)
- **Tools :** Git, GitHub, Postman
- **Core CS Concepts:** Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks

Projects

AI-Powered Health Recommendation System (Java, Spring Boot, PostgreSQL, MongoDB, RabbitMQ) | **GitHub**2026

- Built a backend system to turn raw wearable data across 10 activity types and dynamic custom metrics into personalized, AI-generated health recommendations, instead of just passively logging metrics like typical health apps.
- Designed a 6-service distributed microservices architecture (Config Server, Eureka Service Registry, API Gateway, User/Activity/AI services) with polyglot persistence across 2 databases — PostgreSQL for user identity, MongoDB for health metrics and AI-generated insights.
- Decoupled activity ingestion from AI processing using RabbitMQ async messaging, removing 2-3 seconds of Gemini API latency from the synchronous request path.
- Integrated Keycloak for OAuth2/JWT authentication, validated once at the API Gateway instead of in each of the 3 domain services, centralizing security across the system.

Personal Expense Tracker (Java, Spring Boot, PostgreSQL, JPA, React) | **GitHub** | **Live Demo** 2025

- Built a full-stack personal finance tracker for expenses, savings, and debts, replacing scattered spreadsheets with a unified dashboard.
- Implemented 32 REST API endpoints across 8 resource types (auth, expenses, categories, savings, debts, bills, subscriptions) using Spring Boot and JPA.
- Designed a backend with isolated data access, validated with 9 test profiles and 160+ records.

Experience

Software Developer Intern, Senraa / Evaao Group

Oct 2025 – Apr 2026

- Integrated computer vision tools into an AR-based virtual try-on product, covering image overlay, silhouette detection, and live camera feed processing.
- Built a weighted image scoring system (silhouette framing, brightness, sharpness, posture) that scores input quality out of 100 and rejects images scoring below 70, improving pipeline reliability.
- Evaluated CatVTON on 100+ images across different silhouettes for garment fitting quality, and used IDM-VTON for mask generation; separately tested the PARE model on 10 videos for 3D human mesh generation as a potential improvement to fitting accuracy.

Activities & Achievements

- Finalist – IBM Prarambh Hack Challenge 2025 (Top among 1,200+ teams).
- Certified in Spring 6 and Spring Boot 3 (Telusko).
- Solved 500+ DSA problems on LeetCode and HackerRank; CodeChef rating 1400+.